CrowdSense
Real-time crowd formation & risk assests

Introduction

This project explores a two-stage approach to understanding and managing crowds from CCTV-like video. First, we detect and quantify human gatherings indirectly, using a state-of-the-art density map model to produce head points, counts, and zone-level crowd signals, rather than relying on fragile person detection. Second, we use the temporal evolution of these signals - density, growth, compaction, and flow near bottlenecks- to estimate whether an existing gathering is drifting toward hazardous conditions in the near future. Together, these stages aim to turn raw video streams into interpretable early-warning signals that can help operators spot dangerous crowd build-up before it turns into an incident.

Research Question

Can we detect when a crowd gathering forms in CCTV video and, from its early visual patterns over time, forecast whether it will become high-risk (for example, risk of crowd crush due to overcrowding and violence) within the next X minutes

Literature Review and Existing Approaches

Inspiration and Background

Detecting human gatherings in public spaces is a crucial capability for urban safety, public health, and event monitoring, particularly in dense city environments. Traditional methods for identifying gatherings often rely on explicit detection and tracking of individuals, a task made difficult in real-world scenarios where people are partially occluded, appear at varying scales, or blend into complex urban backdrops. To overcome these challenges, modern computer vision has shifted toward **density-based crowd analysis**, which treats people not as discrete objects to detect but as contributors to a continuous density distribution over the scene [2]. This perspective allows us to estimate how many people are present and where they are concentrated, even when visual conditions are challenging, and is exemplified by recent density map models such as CSRNet [2] and DM-Count [1] that produce high-quality density fields in congested scenes.

However, in many tragic crowd incidents, such as crushes at festivals, stadiums, or religious events, the core problem is not just that a gathering exists, but that a normal crowd silently evolves into a dangerous one without timely warning [3]. Once we can detect and quantify gatherings in the scene, the next challenge is to understand how those gatherings change over time and whether they are drifting toward hazardous conditions. Instead of relying only on static snapshots, **risk escalation forecasting** looks at how density, movement, and compaction evolve - whether people are packing closer together, slowing down near bottlenecks, or being unable to escape the area, following insights from empirical studies of crowd disasters and congestion metrics that jointly consider density and flow [3,4]. By turning these temporal patterns into an early warning signal, the system aims to forecast danger a few minutes in advance, giving operators a chance to intervene before a dense crowd crosses the line into a hazardous situation, in line with recent work that uses learned crowd features (e.g., density and severity levels) to predict critical states and anomalies in advance [5].

Positioning of This Research

This project builds on advances in crowd counting via **density map regression**, a line of research that bypasses hard detection in favor of learning a smooth, continuous representation of human presence in each frame [2]. Inspired by this perspective, the work adopts **DM-Count** (Distribution Matching for Crowd Counting), a state-of-the-art density map model that offers both high accuracy and well-localized peaks without requiring explicit head bounding boxes during training [1]. By emphasizing **distribution matching** rather than pixel-wise similarity, DM-Count learns from sparse point annotations and generates precise, interpretable density maps [1]. These maps serve as a proxy for physical presence, from which head positions and, subsequently, gatherings can be inferred through spatial aggregation over time, building on a broader family of density-based models such as CSRNet that have demonstrated high-quality density estimation in congested scenes [2]. This density-driven approach is particularly aligned with smart city surveillance goals, where dynamic crowd behavior and limited per-person visibility demand robust, indirect estimation techniques.

On top of this density-based representation, the project extends its scope from *detecting and **quantifying*** gatherings to ***forecasting*** when they may become hazardous. Rather than treating each frame independently, the system uses the time series of density-derived signals, such as people per square meter, density growth rate near bottlenecks, compaction of clusters, and indicators of flow or stagnation, guided by decades of crowd safety research on critical density, stop-and-go waves, and turbulence in large gatherings [3]. Video-based congestion measures that explicitly couple density with motion, such as the **crowd congestion level** introduced for risk assessment in surveillance footage, further motivate the use of joint density-flow indicators for risk modeling [4]. Recent work on simulated crowd data shows that these types of features can drive both **unsupervised anomaly detectors** and **supervised sequence models** for early warning of hazardous states, by learning how high density, low speed, and directional conflicts precede dangerous crowd conditions [5]. In this way, the project is positioned at the intersection of modern computer vision and crowd safety: it leverages state-of-the-art density maps to obtain reliable, geometry-aware crowd measurements, and then applies risk-escalation modeling to transform those measurements into **interpretable early-warning signals** to support real-time decision-making in urban environments.

Prior Work Relevant to Our Approach

Density-Based Crowd Counting

Modern crowd counting predominantly relies on density map regression, where CNNs learn to predict a continuous density field instead of explicit detections. CSRNet established a strong baseline for highly congested scenes by showing that dilated convolutions can produce accurate density estimates in complex urban imagery [2]. DM-Count advanced this line by framing learning as distribution matching between predicted density and point annotations, yielding sharper, better-localized maps and state-of-the-art count accuracy across several benchmarks [1]. Our work adopts this density-regression paradigm and specifically builds on DM-Count's ability to produce high-quality density fields that are suitable not only for counting but also for extracting head positions and zone-level signals.

From Density Maps to Gatherings

Several density-based methods implicitly support localization and hotspot reasoning by treating density peaks and local maxima as proxies for individual heads or crowd clusters [1,2]. This enables downstream tasks such as identifying crowded subregions or estimating per-region occupancy, even when explicit detection is unreliable. Our approach formalizes this idea into a simple, reusable interface: per-frame head points, total count, and zone-level counts, which together provide an operational definition of "gatherings" without introducing a separate gathering detection network.

Crowd Risk, Congestion, and Safety Indicators

Research on crowd disasters and safety has highlighted how dangerous conditions emerge when high density combines with restricted movement and geometric constraints. Helbing et al. showed that extreme densities in confined geometries can lead to stop-and-go waves and turbulent crowd motion, which often precede crush incidents [3]. Bek and Monari proposed a video-based "crowd congestion level" that explicitly couples local density with reduced motion to quantify congestion risk in surveillance footage [4]. These works motivate our choice of risk indicators, density per area, growth near bottlenecks, compaction, and low flow, as physically meaningful signals derived from video.

Learning-Based Crowd Anomaly and Risk Prediction

Recent work has explored learning-based models that operate on crowd-level features to detect or predict hazardous conditions. The SIMCD framework demonstrates how simulated crowd data with labels for "severity" and abnormal states can be used to train both **unsupervised anomaly detectors** (e.g., one-class models) and **supervised sequence models** (e.g., LSTMs) for early warning of dangerous crowd configurations [5]. This supports the idea that short histories of features such as density, speed, and directional consistency are sufficient to forecast risk escalation. In our setting, the same density-derived, geometry-aware features can feed **unsupervised models** like **one-class SVM** (and related anomaly detectors) to learn the envelope of normal crowd behavior from non-incident periods, alongside **rule-based baselines** and, where appropriate, supervised temporal models. Across all these variants, we emphasize keeping the resulting risk scores and their drivers interpretable, so that alerts can be traced back to concrete signals such as rising density, low flow, or unusual motion patterns.

Our Approach: Key Differences

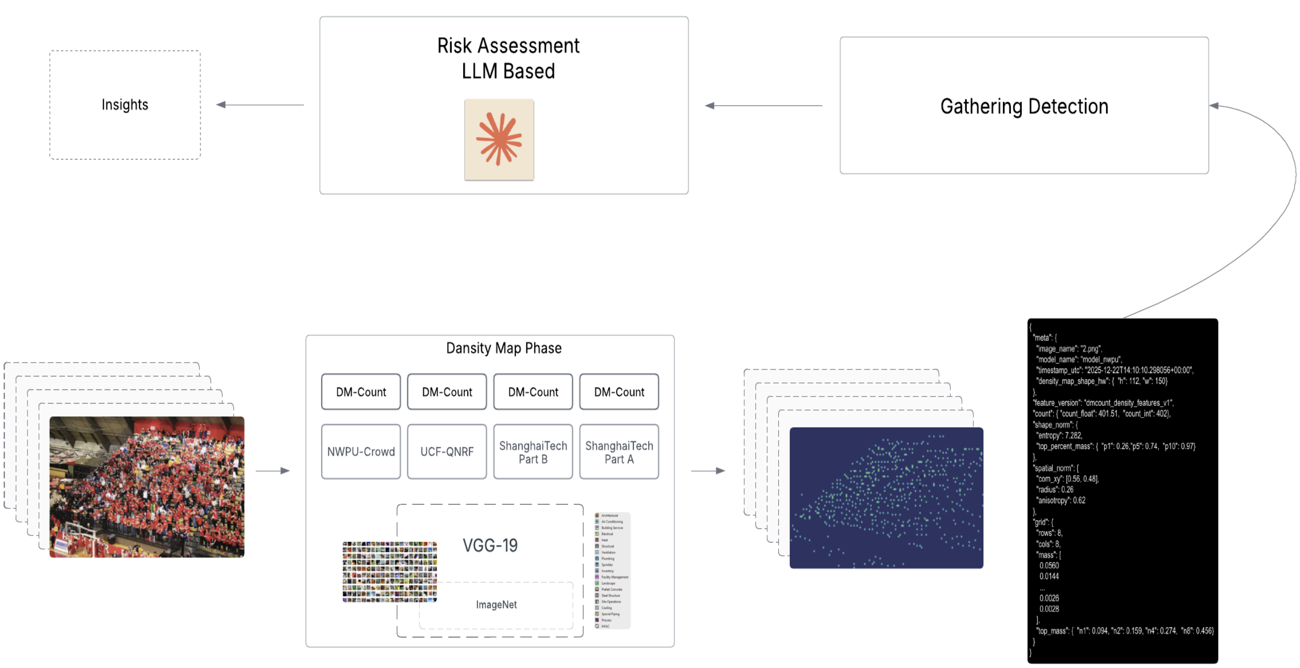
In summary, prior work has shown how to:

- obtain reliable density maps from single frames [1,2]
- identify critical density-flow patterns associated with disasters [3,4]
- use crowd-level features to detect or predict anomalous crowd states [5].

Our contribution is to connect these strands into a two-layer architecture: first, a state-of-the-art density model producing standardized, geometry-aware gathering signals; second, a risk escalation layer that transforms these signals into zone-level risk scores and human-readable drivers. This positions the system as a bridge between advanced crowd counting models and operational, interpretable early-warning tools for urban surveillance.

Methodology

High-Level Architecture



CrowdSense Pipeline Steps

The CrowdSense follows an end-to-end **Sense-Detect-Predict** pipeline to address our research question: “Can we detect when a crowd gathering forms in pictures and from its early visual patterns, forecast whether it will become high risk within the next few minutes?”

It does this by first converting raw imagery into a **spatial crowd representation** (a density map), then compressing that signal into **interpretable features** that capture both size and concentration. Those features support higher-level inference: recognizing a **gathering event** and (when temporal cues are available) estimating whether the crowd is **escalating toward risk**.

From Image To Density Map (Sense → Detect)

CrowdSense starts by sensing the scene through a single input image. The system validates the image, preprocesses it into a model-ready tensor, and then runs a crowd-counting model (DM-Count) to detect people's presence spatially by generating a density map (a heat-like grid where higher values mean more people in that region).

From Density Map To Feature (Detect → Summarize)

Next, the density map is summarized into features: a compact vector of statistics that captures both “how many” and “how they’re distributed.” This includes total count (float/int), concentration and uniformity (entropy, top-percent mass), spatial shape (center of mass, radius, anisotropy), and coarse grid-based distribution signals. In parallel, the density map can be visualized as a heatmap for inspection.

From Feature to Crowd Gathering (Summarize → Detect Gathering)

Then (conceptually, not yet implemented), the system interprets these features to decide whether the scene represents a crowd gathering rather than scattered individuals. This step turns raw “density features” into a gathering state by emphasizing signals like hotspot dominance, spread vs. clustering, and (if available) cluster-level properties derived from the density map. In the paper’s framing, this is where the pipeline transitions from “detecting people” to “detecting a crowd event.”

From Crowd Gathering to Risky Crowd (Predict)

Finally (also conceptual), CrowdSense predicts risk by taking the crowd gathering state and its extracted **JSON attributes** and using an **LLM-based reasoning layer** to turn them into actionable insights: “risky” vs “not risky,” plus *why*.

Instead of only fixed thresholds or classical anomaly models, the LLM interprets the full feature context (counts, concentration, spread, hotspots, clustering signals, and-if available, temporal change like growth/motion) and produces a **risk assessment** + short explanation. The output is a **risk score**/label/alert along with the main drivers it used from the JSON

References

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